

Asian American Journal of Psychology

Racial Discrimination, Distress, Coping Motives, and Alcohol-Related Problems Among U.S.-Born Asian American Young Adults

Derek Kenji Iwamoto, Jeremy C. Kane, Nalini Junko Negi, Anahi Collado, and Davood Tofighi

Online First Publication, January 10, 2022. <http://dx.doi.org/10.1037/aap0000238>

CITATION

Iwamoto, D. K., Kane, J. C., Negi, N. J., Collado, A., & Tofighi, D. (2022, January 10). Racial Discrimination, Distress, Coping Motives, and Alcohol-Related Problems Among U.S.-Born Asian American Young Adults. *Asian American Journal of Psychology*. Advance online publication. <http://dx.doi.org/10.1037/aap0000238>

Racial Discrimination, Distress, Coping Motives, and Alcohol-Related Problems Among U.S.-Born Asian American Young Adults

Derek Kenji Iwamoto¹, Jeremy C. Kane², Nalini Junko Negi³, Anahi Collado⁴, and Davood Tofighi⁵

¹ Department of Psychology, University of Maryland College Park

² Johns Hopkins Bloomberg School of Public Health, New York, United States

³ School of Social Work, University of Maryland Baltimore

⁴ Cofrin Logan Center for Addiction Research and Treatment, Kansas University

⁵ Department of Psychology, University of New Mexico

There is evidence that alcohol use and problem drinking are increasing among Asian Americans. This trend appears to be salient for U.S.-born Asian Americans in particular. Some studies suggest that U.S.-born Asian Americans engage in similar rates of alcohol use compared to other racial groups. Therefore, it is critical to identify the mechanisms of risk and factors associated with alcohol-related problems among this health disparities group. Racial discrimination is considered to be a highly stressful life experience and is significantly associated with alcohol use among ethnic minorities in general. We investigated the role of psychological distress and the drinking to cope motive as underlying mediators between racial discrimination and alcohol-related problems among 1,432 U.S.-born Asian American young adult college students from Southern California. Asian American students reported high rates of psychological distress and instances of racial discrimination. The final structural equation model provided strong fit to the data, and the variables in the model accounted for 44% of the variance in alcohol-related problems. Controlling for alcohol use, Greek fraternity and sorority status, and gender, racial discrimination was significantly associated with alcohol-related problems. Drinking to cope and psychological distress partially mediated the discrimination and alcohol-related problems association. Racial discrimination appears to be an important risk factor for alcohol-related problems among Asian American young adults. This study was a novel investigation into how psychological distress and drinking to cope mechanisms help explain and provide specificity to the racial discrimination and alcohol-related problems association among Asian Americans.

What is the public significance of this article?

U.S.-born Asian American young adults reported frequent incidences of racial discrimination, and significant psychological distress. Racial discrimination and distress increased the risk of reporting alcohol-related problems.

Keywords: Asian Americans, discrimination, alcohol-related problems, drinking to cope, college students

College students have the highest prevalence of alcohol use disorders relative to any other age group (Grucza et al., 2009). National epidemiological studies suggest that over 40% of college students who consume alcohol can be categorized as moderate to heavy drinkers (Johnston et al., 2012). Given the numerous alcohol-related problems associated with heavy alcohol use including relationship problems, drinking and driving, and violence

(Hingson et al., 2017), it is essential to identify the determinants and mechanisms that heighten alcohol-related problems among this highest risk group. While the vast majority of studies have focused on high-risk drinking among white American college students, much less is known about the socially derived stressors and psychosocial factors of drinking among ethnic/racial minority college students, including Asian Americans.

Asian Americans represent the fastest-growing group in the United States, yet this group is often overlooked in the alcohol literature (Hahm et al., 2004; Kim et al., 2014). The “model minority” stereotype, or the notion that Asian Americans are financially well-off, highly educated, and psychologically well-adjusted, is hypothesized to be one reason why Asian American college students remain underrepresented in studies investigating alcohol use among this population. In addition, research studies that aggregate Asian Americans as a whole often obscure significant within-group differences in alcohol use among this group (Iwamoto, Kaya, et al., 2016). While national rates of alcohol use among Asian Americans as a group are low, recent studies suggest that drinking patterns vary significantly among Asian Americans. High-risk drinking and

Derek Kenji Iwamoto  <https://orcid.org/0000-0002-8587-9389>

Jeremy C. Kane  <https://orcid.org/0000-0002-6598-3840>

Nalini Junko Negi  <https://orcid.org/0000-0002-9016-9170>

Davood Tofighi  <https://orcid.org/0000-0001-8523-7776>

Jeremy C. Kane is now at Columbia University, Mailman School of Public Health.

Correspondence concerning this article should be addressed to Derek Kenji Iwamoto, Department of Psychology, University of Maryland College Park, 2147A Biology-Psychology Building, College Park, MD 20742, United States. Email: diwamoto@umd.edu

alcohol use disorders have significantly increased among young adult Asian Americans between 2001–2002 and 2012–2013 (Grant et al., 2017). Other studies have identified subgroups of Asian American young adults who have comparable rates of heavy drinking and alcohol-related problems compared to other racial groups such as white Americans (Cook et al., 2013; Iwamoto et al., 2010). Nativity or whether an individual was born in or outside of the U.S., appears to be one important driver of within-group differences in drinking patterns among this population (Cook et al., 2015; Iwamoto et al., 2010). Asian Americans born in the U.S. report higher alcohol consumption compared to foreign-born Asians (Cook et al., 2013; Iwamoto, Grivel, et al., 2016; Lum et al., 2009). Such differences in drinking behaviors across nativity can be attributed to a complex process of acculturation that may influence drinking behaviors (Lui & Zamboanga, 2018). Regardless, the research is clear that the sociocultural experiences of those who are U.S.-born are distinct from immigrants and aggregate data is likely to mask not only prevalence but also obfuscate unique health risk factors associated with nativity (Miller, 2010).

Racial discrimination is a frequent stressor experienced by Asian American college students and may have especially damaging consequences during this development period (Hall et al., 2015; Tran & Sangalang, 2016). Comprehensive reviews highlight the deleterious effects of racial discrimination on a wide range of health outcomes, including alcohol misuse, psychological distress, depression, and physical health problems (Pascoe & Smart Richman, 2009; Williams & Mohammed, 2009). Yet, Asian Americans have been underrepresented in empirical research on discrimination (Tran & Sangalang, 2016) and even fewer studies have examined the discrimination and drinking relationship among this group (Gilbert & Zemore, 2016). The paucity of studies that have included racial discrimination has found a direct association between discrimination and alcohol use and related problems among community samples of Asian Americans (Chae et al., 2008; Gee et al., 2007; Yoo et al., 2010). However, the relationship between racial discrimination and alcohol use has been relatively untested among U.S.-born Asian American college students despite reports of rising discrimination against Asian Americans (Tessler et al., 2020). This is a significant gap in the literature as discrimination “gets under the skin” heightening risk for alcohol-related problems (Hatzenbuehler, 2009; Hatzenbuehler et al., 2011).

Alcohol use may be driven by an individuals’ motivation and efforts to escape or avoid feelings of distress (Baker et al., 2004; Kenney et al., 2015). Negative reinforcement models suggest that if individuals are overwhelmed by the distress they are experiencing, these individuals may consume alcohol to alleviate this distress (Solomon & Corbit, 1974). While, the positive relationship between psychological distress and alcohol-related problems is well-established (Merrill et al., 2014), the motive of drinking to cope may be a more proximal predictor of alcohol-related problems (Cooper et al., 2016; Rice & Van Arsdale, 2010). Specifically, motivational models, suggest that distress is strongly associated with drinking motives (Kenney et al., 2015). College students who drink to cope with distress have been found to be at an increased risk of experiencing alcohol-related problems in comparison to those who drink for reasons consistent with positive reinforcement (e.g., social purposes or enhancement motives) even after controlling for alcohol consumption (Cox et al., 2006). While several studies have reported significant associations between psychological distress and

alcohol use among Asian Americans (Cheng et al., 2012; Kim et al., 2014), fewer have investigated the role of severe, unavoidable, and uncontrollable stressors (such as, racism) that may confer risk for elevated levels of alcohol-related problems (Chae et al., 2008).

The theoretical mechanisms of risk (i.e., drinking to cope) between racial discrimination and alcohol-related problems among white and ethnic/racial minority college students were first tested by Hatzenbuehler et al. (2011). Coping motives mediated the association between discrimination and alcohol-related problems. The study suggested that college students’ discrimination was indirectly linked to alcohol-related problems through negative affect and drinking to cope for both white and ethnic/racial minority students; however, this study did not disaggregate ethnic/racial minority groups. To our knowledge, there have been no studies that have examined the degree to which racial discrimination has an indirect effect on alcohol-related problems through multiple mechanisms of risk including, psychological distress and drinking to cope, among U.S.-born Asian Americans. One study examined how discrimination was indirectly associated with alcohol-related problems through drinking to cope among underage (under the legal drinking age; 18-year olds) Asian American college students (Le & Iwamoto, 2019). While this study has been formative in advancing understanding regarding the important role of racial discrimination in alcohol use, this study did not include psychological distress and was conducted with a relatively small sample (i.e., $N = 311$) of underage U.S.-born Asian American college students. To this end, the degree to which racial discrimination may indirectly operate through psychological distress which, in turn, may be associated with drinking to cope and alcohol-related problems, is still unknown. This gap is significant as there is evidence to believe that the indirect effect of racial discrimination on alcohol-related problems may be stronger through psychological distress (Huynh et al., 2014; Syed & Juan, 2012). Studying these distinct relationships with a large sample of underage and legal-drinking age Asian American young adults provides us with a broader understanding of underlining mechanisms of risk with implications for informing targeted interventions and programming with this group.

Based on the negative reinforcement and motivational theoretical models and related empirical findings (Cooper et al., 2016; Kenney et al., 2015; Solomon & Corbit, 1974), we hypothesize that higher reported levels of racial discrimination, distress, and drinking to cope will be significantly associated with alcohol-related problems. Second, we test the degree to which discrimination may have an indirect effect on alcohol-related problems through two mechanisms of risk: Psychological distress and drinking to cope. We hypothesize that discrimination will heighten distress, which will in turn relate to drinking to cope with motives, which will result in more alcohol-related problems.

Method

Participants and Procedures

The current study draws upon unique secondary data that has not been published from the larger study (author). Participants in the current study were 1,432 Asian American undergraduates from a large public university located in southern California. Chinese Americans consisted of the largest ethnic group (32.1%), followed by Vietnamese Americans (17.5%), Filipino Americans (14.6%),

Korean Americans (11.5%), Other Asian ethnicity (e.g., Taiwanese, Hmong, Cambodian; 9.1%), Indian American (8.1%), and Japanese American (7.1%). The average age of the participants was 19.81 years ($SD = 1.36$; age range 18–24), and 29.2% identified as seniors, 27.1% as juniors, 21.7% as sophomores, and 22% as freshmen. The sample was predominantly women (73.2%), and 23.3% were involved in a Greek sorority/fraternity. The percentage of ethnic groups represented in the current sample was reflective of the university's Asian ethnic group demographic makeup. The institutional review board approved all procedures in the study. Participants were recruited using an online forum through which students from all majors (e.g., psychology, engineering, chemistry, social ecology, and humanities) can participate in experiments for extra credit. In order to participate in the study, participants had to identify as Asian American and born in the U.S.

Measures

Demographics

The demographics section assessed age, ethnicity, nativity (e.g., yourself or your sibling were the first born in the United States), and Greek affiliation (e.g., fraternity or sorority).

Everyday Racial Discrimination (Williams et al., 1997)

The Everyday Racial Discrimination (ERD) measure is widely used and assesses the frequency of unfair treatment in the day-to-day lives of participants due to race/ethnicity. The stem of the items includes, "Based on your race": Being treated with less courtesy than others; being treated with less respect than others; receiving poorer service than others in restaurants or stores; and people acting as if you are not smart. The scale has a total of nine items that measure the frequency of such experiences. Each item is rated on a 6-point scale ranging from: *never, less than once a year, a few times a year, a few times a month, at least once a week, and almost every day*. The scale has been validated and used in epidemiological studies with Asian American community samples (Gee et al., 2007, 2009). An overall composite score for the racial discrimination measures was calculated by summing participants' responses on the nine items. The reliability estimate for the racial discrimination measure in the current study was $\alpha = .90$.

Kessler Psychological Distress Scale (Kessler et al., 2002)

The Kessler Psychological Distress Scale (K-10) is a widely used and well-validated measure of distress caused by anxiety and depressive feelings. All items are answered in the context of how a participant felt in the past 30 days. Sample items include, "... about how often did you feel tired for no good reason?" or "... how often did you feel nervous?" Responses options are on a 5-point Likert-type scale, ranging from 1 (*none of the time*) to 5 (*all of the time*). Scores for the K-10 were assessed by summing the responses for each individual, with a possible score range 10–50. Scores falling between 20 and 24 indicate "mild mental disorder," scores between 25 and 29 indicate potential for "moderate mental disorder," and scores above 30 suggest the likelihood of a "severe mental disorder" (Kessler et al., 2002). Convergent validity evidence for the K10 clinical cut-off scores was demonstrated through

significant correlations with serious mental illness and Diagnostic Statistical Manual disorders (Kessler et al., 2002, 2003, 2010; Spies et al., 2009). The present study's reliability estimate for the K-10 was $\alpha = .91$.

Drinking to Cope

The five-item drinking to cope motives scale is from the Coping and Enhancement Motives scale developed by Cooper (1994) and is the most widely used coping motives measure (Kuntsche et al., 2005). This scale assesses the degree to which an individual uses alcohol to "escape, avoid, or otherwise regulate negative emotions" (Cooper et al., 1995, p.991). A sample item includes "How often do you drink to forget your worries?" It has a total of four items and is scored on a 6-point Likert scale that ranges from 1 (*almost never/never*) to 5 (*almost always/always*). All five items are summed up to provide an overall composite score of drinking to cope. Internal consistency for the drinking to cope measure in this current study was $\alpha = .92$.

Quantity of Alcohol

Quantity was assessed by summing the number of drinks consumed on an average week in the past 3 months. Participants were asked to approximate their typical alcohol consumption for each day of the week over the past 90 days, similar to the gold-standard Timeline Followback interview assessment (Sobell & Sobell, 1992).

Rutgers Alcohol Problems Index (White & Labouvie, 1989)

The Rutgers Alcohol Problems Index (RAPI) examines the frequency at which an individual experienced alcohol-related problems or negative consequences while or because of drinking over the past year. This well-validated instrument assesses the occurrence of a total of 23 negative consequences and response options are on a 5-point Likert scale: 0 (*never*) to 4 (*10 or more times*). Sample items include "Not able to work or study for a test" and "Missed out on other things because you spend too much money on alcohol." The current study used the three distinct subscales on the RAPI derived by Martens and colleagues (Martens et al., 2007, 2008) which were labeled Abuse/Dependence (12 items), Individual Consequences (seven items), and Social Consequences (four items). Scores for the RAPI subscales were derived by summing the responses for each individual based on the subscale. The subscales were used as indicators/parcels of the latent factor of alcohol-related problems. The internal consistency estimate of the three subscales was $\alpha = .87, .84, .82$ for Abuse/Dependence, Individual Consequences, and Social Consequences, respectively.

Data Analytic Plan

Structural equation modeling (SEM) was the primary data analytic approach used in the current study. SEM was performed using *Mplus 7* (Muthen & Muthen, 2012). SEM can test and model complex patterns of relationships between latent factors, while simultaneously test multiple mediators and outcome variables. Based on the procedures outlined by Matsunaga (2008), we created item parcels as indicators of latent variables (racial discrimination and psychological distress). Individual items from the discrimination and

K10 measures were ranked ordered by factor loadings, and three parcels were created for each factor. There are several advantages of creating item parcels. Item parcels generally have higher reliability estimates than single items thus reducing measurement error and result in better fitting models (Coffman & MacCallum, 2005). Findings from simulation studies suggest that item parcels reduce the number of parameters in the model thus significantly reducing parameter bias (Bandalos, 2002; Rhemtulla, 2016). Item parceling also increases the power to detect model misspecification and provides more accurate parameter estimates. Since the confirmatory factor analyses (CFA) measurement model of the drinking to cope motive is well-documented (Gilson et al., 2013; Nehlin & Öster, 2019), we used all the observed indicators for the drinking to cope measure. Following Martens et al. (2007) recommendations, the alcohol-related problems latent factor was created using alcohol abuse, personal problems, and social problems subscales. The comparative fit index (CFI), Tucker-Lewis index (TLI), and root mean square error of approximation (RMSEA) were used to assess the fit of the measurement and structural models. Conventional standards suggest that CFI and TLI values greater than .95 and RMSEA values of .05 indicate near model-to-data fit (Hu & Bentler, 1999), with RMSEA values of .08 suggesting fair fit (Brown & Cudeck, 1993). Full information maximum likelihood (FIML) procedures were used to handle missing data. FIML procedures do not input data but rather through Expectation-maximization (EM) procedure, estimates the standards and parameters in the model (Enders, 2001). To test for mediation, the bootstrap bias-corrected standard errors were performed using 1,000 bootstraps (Mallinckrodt et al., 2006).

Results

Descriptive Data

Descriptive statistics were conducted and the average distress score was 21.96 ($SD = 7.25$), which indicates that, on average, the scores fall within the clinical range (e.g., scores between 20 and 24 indicate a mild disorder; Kessler et al., 2002, 2003, 2010). Thirty-percent of the sample scored 25 or higher, indicating high probability of detecting a moderate to severe mental illness. All the participants reported at least one incident of racial discrimination (e.g., participants reported on at least one item of the measure that they experienced racial discrimination), and 59% reported at least two incidents of racial discrimination (e.g., participants endorsed at least two items on the measure). The correlation analysis revealed that racial discrimination was positively associated with psychological distress ($r = .41, p < .001$), drinking to cope ($r = .20, p < .001$), and alcohol-related problems ($r = .22, p < .001$). Psychological distress was positively related to drinking to cope ($r = .29, p < .001$) and alcohol-related problems ($r = .25, p < .001$), and drinking to cope was related to alcohol-related problems ($r = .62, p < .001$).

Measurement Model

The latent constructs in the measurement model, included discrimination, psychological distress, drinking to cope, and alcohol-related problems, were allowed to covary and provided adequate fit to the data $\chi^2(71) = 281.69, p < .001$, CFI = .99, TLI = .98, and RMSEA = .046.

Structural Model

We tested the structural model with all the observed variables (the covariates, alcohol consumption, and Greek status) included in the model, which resulted in the structural model offering a good fit to the data, $\chi^2(133) = 15,375, p < .001$, CFI = .96, TLI = .95 and RMSEA = .062 (CI = .057–.066). The significant pathways between factors and the *R*-square for psychological distress, drinking to cope, and alcohol-related problems latent factors are reported in Figure 1. Racial discrimination was significantly associated with higher psychological distress ($\beta = .45, p < .001$) and psychological distress was positively related to drinking to cope ($\beta = .32, p < .001$). Controlling for alcohol consumption ($\beta = .32, p < .001$), Greek status ($\beta = .08, p < .001$), and gender ($\beta = .05, p = .07$), higher reports of racial discrimination ($\beta = .08, p < .001$), psychological distress ($\beta = .06, p < .05$), and drinking to cope ($\beta = .51, p < .001$), were positively associated with more alcohol-related problems (results are displayed in Figure 1). The bootstrap bias-corrected standard errors, which included 1,000 bootstrapped samples tests of indirect effects, detected two indirect effects: (a) racial discrimination (.073, $p < .001$, 95% CI [.054, .093]) had an indirect effect on alcohol-related problems through psychological distress and drinking to cope, and (b) racial discrimination had an indirect effect on alcohol-related problems through psychological distress (.03, $p < .05$, 95% CI [.0005, .053]). See Table 1.

Discussion

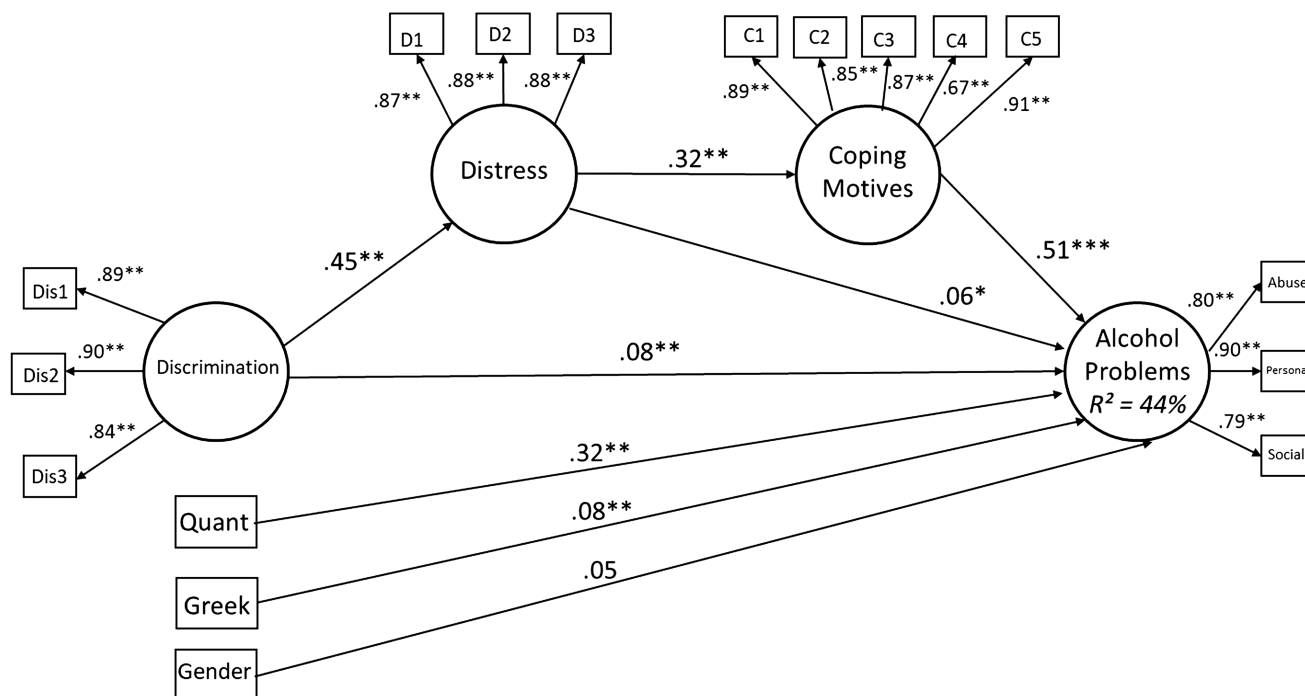
Asian Americans experience frequent racial discrimination yet there has been limited research that has investigated its relationship to alcohol-related problems. This gap in the literature has persisted despite advances in our knowledge about the mediated associations between racial discrimination and alcohol outcomes (Gilbert & Zemore, 2016). The present study significantly advances the scant literature in this area by first, highlighting the direct and indirect associations between racial discrimination and alcohol-related problems, and second, identifying psychological distress and drinking to cope as potential mechanisms of risk. Meaningfully, results reveal that all participants reported experiencing racial discrimination and over 30% scored in the clinical range on the K10 (Kessler et al., 2003).

Controlling for well-established covariates of problematic alcohol use among college students including, Greek status (Borsari et al., 2007), quantity of alcohol consumption and gender, racial discrimination was found to be directly associated with alcohol-related problems. The direct effects of discrimination on alcohol use and alcohol-related problems are consistent with previous studies with community samples of Asian Americans (Chae et al., 2008; Gee et al., 2007; Yoo et al., 2010). The second hypothesis was supported as well, wherein racial discrimination had an indirect relationship on alcohol-related problems through psychological distress and the drinking to cope motive. Current results are consistent with Hatzenbuehler et al.'s (2011) findings with white and ethnic minority college students.

Racial discrimination is one particularly distinct source of stress that has been posited to be associated with psychological distress, and distress has been found to mediate the association between stress and alcohol use (Bujarski & Ray, 2014). Asian American

Figure 1

Structural Equation Model of Racial Discrimination, Distress, Drinking to Cope, Alcohol Consumption, and Alcohol-Related Problems



college students in our study who experienced relatively higher frequencies of racial discrimination were more likely to report more distress and experience greater alcohol-related problems. Based on motivational models (Cooper et al., 2016), it is plausible that racial discrimination heightens psychological distress, thus, these individuals may engage in problematic alcohol use to cope with and regulate their distress due to discrimination. Findings add to growing evidence regarding the distinct and toxic role of racial discrimination on psychological distress and alcohol-related problems

Due to limitations attributable to the cross-sectional design of this study, it is difficult to definitively conclude the temporal direction and influence among the variables in the model (Maxwell & Cole, 2007). A longitudinal design would be better suited to prospectively investigate the prognosis of alcohol-related problems and more accurately approximate the indirect effects of racial discrimination. While previous studies have suggested differences in drinking rate between Asian ethnic groups, we were unable to examine ethnic group differences due to unequal sizes of Asian American ethnic subgroups in our sample. In addition, although we controlled for gender, this sample has a disproportionate ratio of women to men

which is a common phenomenon in research conducted with college students (Lui & Zamboanga, 2018). To this end, future studies should consider having a more balanced sample, including larger and more Asian American ethnic groups (e.g., Filipino, Korean, South Asian Americans), and examine gender-relevant factors and differences across ethnic groups in relation to alcohol outcomes. Cultural orientation and alcohol beliefs may also be other salient factors to examine across Asian ethnic groups and alcohol use (Lui et al., 2020). Although there is strong evidence for the use of the Everyday Discrimination Scale with Asian Americans, a culturally specific measure such as the Subtle and Blatant Racism Scale might provide a more nuanced understanding of the specific types racism experienced by Asian Americans (Yoo, Steger, & Lee, 2010).

Despite these limitations, there are a few notable strengths of the present study. First, the large sample size of U.S.-born Asian American college students provided the opportunity to examine the associations of interest in this underrepresented population. Additionally, this study builds upon previous investigations into the discrimination-alcohol problem relationship by testing the influence of two theoretically important mechanisms of risk, including drinking motive and distress. Despite these limitations, the

Table 1

Indirect Effects of Racial Discrimination on Alcohol-Related Problems Through Psychological Distress, Drinking to Cope Motive

Variable	Indirect effect	SE	Bootstrap 95% CI for mean indirect effect
Racial discrimination → Distress → Problems	.03	.015	[.0005, .053]
Racial discrimination → Distress → Drinking to cope → Problems	.073	.035	[.054, .093]

Note. Bootstrap bias-corrected standard error using 1,000 bootstrapped samples.

current study provides greater specificity regarding the distinct pathways through which discrimination “gets under the skin” (Hatzenbuehler, 2009) and confers risk for the development of alcohol-related problems for an underrepresented and overlooked group in the alcohol literature, U.S.-born Asian Americans. These findings bear significant clinical implications, particularly in consideration of the high prevalence of psychological distress and racial discrimination among our sample. Counter to assumptions based on the model minority stereotype, 30% of Asian Americans sampled scored within the “moderate to severe mental disorder” range on the psychological distress measure (Kessler et al., 2003, 2010). This finding highlights the high clinical levels of severe distress often overlooked among Asian Americans. All of the Asian Americans in the sample reported experiencing racism and 59% reported experiencing two incidents of racism. These findings highlight that, counter to the “model minority” myth, this group is not protected from experiences of racial discrimination. Considering that racial discrimination seems pertinent to problematic drinking among this sample, it is important to assess, address, and validate the client’s racial discrimination experiences in clinical practice. In addition, as Asian Americans are often clinically overlooked with respect to alcohol use disorders (Cheng et al., 2018), efforts to increase assessment are critical to ensure appropriate intervention and treatment. Clinically targeting the relevant mechanisms of risk (e.g., psychological distress and the drinking to cope motive) and providing alternative active coping strategies might also be facilitative in the reduction of alcohol-related problems. Finally, the toxic impact of racial discrimination on the health of Asian Americans must be addressed. Public health campaigns that help change public perception regarding the “model minority” myth could be one critical way to move past its deleterious impact. Psychologists should also work with Asian American antiracist activists to think through and develop antiracism college programming for Asian American students with the goal of minimizing the corrosive impact of racism on distress and alcohol-related problems.

Highlights

- Alcohol use and alcohol-related problems appears to be increasing among Asian Americans.
- U.S.-born Asian Americans are more likely to report drinking at similar rates as other groups.
- Racial discrimination appears to be directly and indirectly associated to alcohol-related problems.
- Majority of the Asian American in the current study reported experiencing racial discrimination and elevated levels of psychological distress.

References

- Baker, T. B., Piper, M. E., McCarthy, D. E., Majeskie, M. R., & Fiore, M. C. (2004). Addiction motivation reformulated: An affective processing model of negative reinforcement. *Psychological Review*, 111(1), 33–51. <https://doi.org/10.1037/0033-295X.111.1.33>
- Bandalos, D. L. (2002). The effects of item parceling on goodness-of-fit and parameter estimate bias in structural equation modeling. *Structural Equation Modeling*, 9, 78–102. https://doi.org/10.1207/S15328007SEM0901_5
- Borsari, B., Murphy, J. G., & Barnett, N. P. (2007). Predictors of alcohol use during the first year of college: Implications for prevention. *Addictive Behaviors*, 32, 2062–2086. <https://doi.org/10.1016/j.addbeh.2007.01.017>
- Brown, M. W., & Cudeck, R. (1993). Alternative ways of assessing model fit. In K. A. Bollen & J. S. Long (Eds.), *Testing structural equation models* (pp. 136–162). Sage Publications.
- Bujarski, S., & Ray, L. A. (2014). Negative affect is associated with alcohol, but not cigarette use in heavy drinking smokers. *Addictive Behaviors*, 39(12), 1723–1729. <https://doi.org/10.1016/j.addbeh.2014.07.019>
- Chae, D. H., Takeuchi, D. T., Barbeau, E. M., Bennett, G. G., Lindsey, J. C., Stoddard, A. M., & Krieger, N. (2008). Alcohol disorders among Asian Americans: Associations with unfair treatment, racial/ethnic discrimination, and ethnic identification (the national Latino and Asian Americans study, 2002–2003). *Journal of Epidemiology and Community Health*, 62, 973–979. <https://doi.org/10.1136/jech.2007.066811>
- Cheng, A. W., Iwamoto, D. K., & McMullen, D. (2018). Model minority stereotype and the diagnosis of alcohol use disorders: Implications for practitioners working with Asian Americans. *Journal of Ethnicity in Substance Abuse*, 17, 255–272. <https://doi.org/10.1080/15332640.2016.1175990>
- Cheng, A. W., Lee, C. S., & Iwamoto, D. K. (2012). Heavy drinking, poor mental health, and substance use among Asian Americans in the NLAAS: A gender-based comparison. *Asian American Journal of Psychology*, 3, 160–167. <https://doi.org/10.1037/a0028306>
- Coffman, D. L., & MacCallum, R. C. (2005). Using parcels to convert path analysis models into latent variable models. *Multivariate Behavioral Research*, 40, 235–259. https://doi.org/10.1207/s15327906mbr4002_4
- Cook, W. K., Bond, J., Karriker-Jaffe, K., & Zeng, S. (2013). Who’s at risk? Ethnic drinking cultures, foreign nativity, and problem drinking among Asian American young adults. *Journal of Studies on Alcohol and Drugs*, 74, 532–541. <https://doi.org/10.15288/jsad.2013.74.532>
- Cook, W. K., Karriker-Jaffe, K. J., Bond, J., & Lui, C. (2015). Asian American problem drinking trajectories during the transition to adulthood: Ethnic drinking cultures and neighborhood contexts. *American Journal of Public Health*, 105(5), 1020–1027. <https://doi.org/10.2105/AJPH.2014.302196>
- Cooper, M. L. (1994). Motivations for alcohol use among adolescents: Development and validation of a four-factor model. *Psychological Assessment*, 6, 117–128. <https://doi.org/10.1037/1040-3590.6.2.117>
- Cooper, M. L., Frone, M. R., Russell, M., & Mudar, P. (1995). Drinking to regulate positive and negative emotions: A motivational model of alcohol use. *Journal of Personality and Social Psychology*, 69, 990–1005. <https://doi.org/10.1037/0022-3514.69.5.990>
- Cooper, M. L., Kuntsche, E., Levitt, A., Barber, L. L., & Wolf, S. (2016). Motivational models of substance use: A review of theory and research on motives for using alcohol, marijuana, and tobacco. In K. J. Sher (Ed.), *The oxford handbook of substance use and substance use disorders* (Vol. 1, pp. 375–421). Oxford University Press.
- Cox, W. M., Hosier, S. G., Crossley, S., Kendall, B., & Roberts, K. L. (2006). Motives for drinking, alcohol consumption, and alcohol-related problems among British secondary-school and university students. *Addictive Behaviors*, 31(12), 2147–2157. <https://doi.org/10.1016/j.addbeh.2006.02.023>
- Enders, C. K. (2001). The performance of the full maximum likelihood estimator in multiple regression models with missing data. *Educational and Psychological Measurement*, 61(5), 713–740. <https://doi.org/10.1177/0013164401615001>
- Gee, G. C., Ro, A., Shariff-Marco, S., & Chae, D. (2009). Racial discrimination and health among Asian Americans: Evidence, assessment, and directions for future research. *Epidemiologic Reviews*, 31, 130–151. <https://doi.org/10.1093/epirev/mxp009>

- Gee, G. C., Spencer, M., Chen, J., Yip, T., & Takeuchi, D. T. (2007). The association between self-reported racial discrimination and 12-month DSM-IV mental disorders among Asian Americans nationwide. *Social Science & Medicine*, 64(10), 1984–1996. <https://doi.org/10.1016/j.socsci.med.2007.02.013>
- Gilbert, P. A., & Zemore, S. E. (2016). Discrimination and drinking: A systematic review of the evidence. *Social Science & Medicine*, 161, 178–194. <https://doi.org/10.1016/j.socscimed.2016.06.009>
- Gilson, K. M., Bryant, C., Bei, B., Komiti, A., Jackson, H., & Judd, F. (2013). Validation of the Drinking Motives Questionnaire (DMQ) in older adults. *Addictive Behaviors*, 38, 2196–2202. <https://doi.org/10.1016/j.addbeh.2013.01.021>
- Grant, B. F., Chou, S. P., Saha, T. D., Pickering, R. P., Kerridge, B. T., Ruan, W. J., Huang, B., Jung, J., Zhang, H., Fan, A., & Hasin, D. S. (2017). Prevalence of 12-month alcohol use, high-risk drinking, and DSM-IV alcohol use disorder in the United States, 2001–2002 to 2012–2013: Results from the National Epidemiologic Survey on Alcohol and Related Conditions. *JAMA Psychiatry*, 74, 911–923. <https://doi.org/10.1001/jama.psychiatry.2017.2161>
- Gruza, R. A., Norberg, K. E., & Bierut, L. J. (2009). Binge drinking among youths and young adults in the United States: 1979–2006. *Journal of the American Academy of Child & Adolescent Psychiatry*, 48(7), 692–702. <https://doi.org/10.1097/CHI.0b013e3181a2b32f>
- Hahm, H. C., Lahiff, M., & Guterman, N. B. (2004). Asian Americans adolescents' acculturation, binge drinking and alcohol and tobacco-using peers. *Journal of Community Psychology*, 32, 295–308. <https://doi.org/10.1002/jcop.20002>
- Hall, M. E., Williams, R. D., Jr., Penhollow, T. M., Rhoads, K. E., & Hunt, B. P. (2015). Factors associated with discrimination among minority college students. *American Journal of Health Behavior*, 39(3), 318–329. <https://doi.org/10.5993/AJHB.39.3.4>
- Hatzenbuehler, M. L. (2009). How does sexual minority stigma “get under the skin”? A psychological mediation framework. *Psychological Bulletin*, 135(5), 707–730. <https://doi.org/10.1037/a0016441>
- Hatzenbuehler, M. L., Corbin, W. R., & Fromme, K. (2011). Discrimination and alcohol-related problems among college students: A prospective examination of mediating effects. *Drug and Alcohol Dependence*, 115(3), 213–220. <https://doi.org/10.1016/j.drugalcdep.2010.11.002>
- Hingson, R., Zha, W., & Smyth, D. (2017). Magnitude and trends in heavy episodic drinking, alcohol-impaired driving, and alcohol-related mortality and overdose hospitalizations among emerging adults of college ages 18–24 in the United States, 1998–2014. *Journal of Studies on Alcohol and Drugs*, 78, 540–548. <https://doi.org/10.15288/jsad.2017.78.540>
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6, 1–55. <https://doi.org/10.1080/10705519909540118>
- Huynh, Q. L., Devos, T., & Goldberg, R. (2014). The role of ethnic and national identifications in perceived discrimination for Asian Americans: Toward a better understanding of the buffering effect of group identifications on psychological distress. *Asian American Journal of Psychology*, 5, 161–171. <https://doi.org/10.1037/a0031601>
- Iwamoto, D. K., Corbin, W., & Fromme, K. (2010). Trajectory classes of heavy episodic drinking among Asian American college students. *Addiction*, 105, 1912–1920. <https://doi.org/10.1111/j.1360-0443.2010.03019.x>
- Iwamoto, D. K., Grivel, M. M., Cheng, A. W., & Zamboanga, B. L. (2016). Asian American and white college students' heavy episodic drinking behaviors and alcohol-related problems. *Substance Use & Misuse*, 51, 1384–1392. <https://doi.org/10.3109/10826084.2016.1170142>
- Iwamoto, D. K., Kaya, A., Grivel, M., & Clinton, L. (2016). Under-researched demographics: Heavy episodic drinking and alcohol-related problems among Asian Americans. *Alcohol Research: Current Reviews*, 38(1), 17–25.
- Johnston, L. D., O'Malley, P. M., Bachman, J. G., & Schulenberg, J. E. (2012). *Monitoring the future national survey results on drug use, 1975–2011* (Vol. 2). Institute for Social Research, University of Michigan.
- Kenney, S., Jones, R. N., & Barnett, N. P. (2015). Gender differences in the effect of depressive symptoms on prospective alcohol expectancies, coping motives, and alcohol outcomes in the first year of college. *Journal of Youth and Adolescence*, 44, 1884–1897. <https://doi.org/10.1007/s10964-015-0311-3>
- Kessler, R. C., Andrews, G., Colpe, L. J., Hiripi, E., Mroczek, D. K., Normand, S.-L. T., Walters, E. E., & Zaslavsky, A. M. (2002). Short screening scales to monitor population prevalences and trends in non-specific psychological distress. *Psychological Medicine*, 32(6), 959–976. <https://doi.org/10.1017/S0033291702006074>
- Kessler, R. C., Barker, P. R., Colpe, L. J., Epstein, J. F., Gfroerer, J. C., Hiripi, E., Howes, M. J., Normand, S. L. T., Manderscheid, R. W., Walters, E. E., & Zaslavsky, A. M. (2003). Screening for serious mental illness in the general population. *Archives of General Psychiatry*, 60(2), 184–189. <https://doi.org/10.1001/archpsyc.60.2.184>
- Kessler, R. C., Green, J. G., Gruber, M. J., Sampson, N. A., Bromet, E., Cuitan, M., Furukawa, T. A., Gureje, O., Hinkov, H., Hu, C. Y., Lara, C., Lee, S., Mneimneh, Z., Myer, L., Oakley-Browne, M., Posada-Villa, J., Sagar, R., Viana, M. C., & Zaslavsky, A. M. (2010). Screening for serious mental illness in the general population with the K6 screening scale: Results from the WHO World Mental Health (WMH) survey initiative. *International Journal of Methods in Psychiatric Research*, 19(Suppl 1), 4–22. <https://doi.org/10.1002/mpr.310>
- Kim, S. S., Lee, H., Kiang, P., Kalman, D., & Ziedonis, D. M. (2014). Factors associated with alcohol problems among Asian American college students: Gender, ethnicity, smoking and depressed mood. *Journal of Substance Use*, 19, 12–17. <https://doi.org/10.3109/14659891.2012.709912>
- Kuntsche, E., Knibbe, R., Gmel, G., & Engels, R. (2005). Why do young people drink? A review of drinking motives. *Clinical Psychology Review*, 25, 841–861. <https://doi.org/10.1016/j.cpr.2005.06.002>
- Le, T. P., & Iwamoto, D. K. (2019). A longitudinal investigation of racial discrimination, drinking to cope, and alcohol-related problems among underage Asian American college students. *Psychology of Addictive Behaviors*, 33(6), 520–528. <https://doi.org/10.1037/adb0000501>
- Lui, P. P., Berkley, S. R., & Zamboanga, B. L. (2020). College alcohol belief and alcohol use: Testing moderations by cultural orientations and ethnicity. *Journal of Counseling Psychology*, 67, 184–194. <https://doi.org/10.1037/cou0000374>
- Lui, P. P., & Zamboanga, B. L. (2018). Acculturation and alcohol use among Asian Americans: A meta-analytic review. *Psychology of Addictive Behaviors*, 32, 173–186. <https://doi.org/10.1037/adb0000340>
- Lum, C., Corliss, H. L., Mays, V. M., Cochran, S. D., & Lui, C. K. (2009). Differences in the drinking behaviors of Chinese, Filipino, Korean, and Vietnamese college students. *Journal of Studies on Alcohol and Drugs*, 70, 568–574. <https://doi.org/10.15288/jsad.2009.70.568>
- Mallinckrodt, B., Abraham, W. T., Wei, M., & Russell, D. W. (2006). Advances in testing the statistical significance of mediation effects. *Journal of Counseling Psychology*, 53, 372–378. <https://doi.org/10.1037/0022-0167.53.3.372>
- Martens, M. P., Martin, J. L., Hatchett, E. S., Fowler, R. M., Fleming, K. M., Karakashian, M. A., & Cimini, M. D. (2008). Protective behavioral strategies and the relationship between depressive symptoms and alcohol-related negative consequences among college students. *Journal of Counseling Psychology*, 55, 535–541. <https://doi.org/10.1037/a0013588>
- Martens, M. P., Neighbors, C., Dams-O'Connor, K., Lee, C. M., & Larimer, M. E. (2007). The factor structure of a dichotomously scored Rutgers Alcohol Problem Index. *Journal of Studies on Alcohol and Drugs*, 68, 597–606. <https://doi.org/10.15288/jsad.2007.68.597>

- Matsunaga, M. (2008). Item parceling in structural equation modeling: A primer. *Communication Methods and Measures*, 2, 260–293. <https://doi.org/10.1080/19312450802458935>
- Maxwell, S. E., & Cole, D. A. (2007). Bias in cross-sectional analyses of longitudinal mediation. *Psychological Methods*, 12, 23–44. <https://doi.org/10.1037/1082-989X.12.1.23>
- Merrill, J. E., Wardell, J. D., & Read, J. P. (2014). Drinking motives in the prospective prediction of unique alcohol-related consequences in college students. *Journal of Studies on Alcohol and Drugs*, 75, 93–102. <https://doi.org/10.15288/jsad.2014.75.93>
- Miller, M. J. (2010). Testing a bilinear domain-specific model of acculturation and enculturation across generational status. *Journal of Counseling Psychology*, 57, 179–186. <https://doi.org/10.1037/a0019089>
- Muthen, L. K., & Muthen, B. O. (2012). *Mplus user's guide* (6th ed.). Muthen & Muthen.
- Nehlin, C., & Öster, C. (2019). Measuring drinking motives in undergraduates: An exploration of the Drinking Motives Questionnaire-Revised in Swedish students. *Substance Abuse Treatment, Prevention, and Policy*, 14, Article 49. <https://doi.org/10.1186/s13011-019-0239-9>
- Pascoe, E. A., & Smart Richman, L. (2009). Perceived discrimination and health: A meta-analytic review. *Psychological Bulletin*, 135, 531–554. <https://doi.org/10.1037/a0016059>
- Rhemtulla, M. (2016). Population performance of SEM parceling strategies under measurement and structural model misspecification. *Psychological Methods*, 21, 348–368. <https://doi.org/10.1037/met0000072>
- Rice, K. G., & Van Arsdale, A. C. (2010). Perfectionism, perceived stress, drinking to cope, and alcohol-related problems among college students. *Journal of Counseling Psychology*, 57(4), 439–450. <https://doi.org/10.1037/a0020221>
- Sobell, L. C., & Sobell, M. B. (1992). Timeline followback: A technique for assessing self-reported ethanol consumption. In J. Allen & R. Z. Litten (Eds.), *Measuring alcohol consumption: Psychosocial and biological methods* (pp. 41–72). Humana Press. https://doi.org/10.1007/978-1-4612-0357-5_3
- Solomon, R. L., & Corbit, J. D. (1974). An opponent-process theory of motivation. I. Temporal dynamics of affect. *Psychological Review*, 81, 119–145. <https://doi.org/10.1037/h0036128>
- Spies, G., Stein, D. J., Roos, A., Faure, S. C., Mostert, J., Seedat, S., & Vythilingum, B. (2009). Validity of the Kessler 10 (K-10) in detecting DSM-IV defined mood and anxiety disorders among pregnant women. *Archives of Women's Mental Health*, 12, 69–74. <https://doi.org/10.1007/s00737-009-0050-0>
- Syed, M., & Juan, M. J. D. (2012). Discrimination and psychological distress: Examining the moderating role of social context in a nationally representative sample of Asian American adults. *Asian American Journal of Psychology*, 3, 104–120. <https://doi.org/10.1037/a0025275>
- Tessler, H., Choi, M., & Kao, G. (2020). The anxiety of being Asian American: Hate crimes and negative biases during the COVID-19 pandemic. *American Journal of Criminal Justice*, 45, 1–11. <https://doi.org/10.1007/s12103-020-09541-5>
- Tran, A. G., & Sangalang, C. C. (2016). Personal discrimination and satisfaction with life: Exploring perceived functional effects of Asian American race/ethnicity as a moderator. *Cultural Diversity and Ethnic Minority Psychology*, 22, 83–92. <https://doi.org/10.1037/cdp0000052>
- White, H. R., & Labouvie, E. W. (1989). Towards the assessment of adolescent problem drinking. *Journal of Studies on Alcohol*, 50, 30–37. <https://doi.org/10.15288/jsa.1989.50.30>
- Williams, D. R., & Mohammed, S. A. (2009). Discrimination and racial disparities in health: Evidence and needed research. *Journal of Behavioral Medicine*, 32, 20–47. <https://doi.org/10.1007/s10865-008-9185-0>
- Williams, D. R., Yu, Y., Jackson, J. S., & Anderson, N. B. (1997). Racial differences in physical and mental health: Socio-economic status, stress and discrimination. *Journal of Health Psychology*, 2, 335–351. <https://doi.org/10.1177/135910539700200305>
- Yoo, H. C., Steger, M. F., & Lee, R. M. (2010). Validation of the subtle and blatant racism scale for Asian American college students (SABR-A²). *Cultural Diversity and Ethnic Minority Psychology*, 16, 323–334. <https://doi.org/10.1037/a0018674>

Received March 9, 2020

Revision received February 14, 2021

Accepted February 25, 2021 ■